How particle shape reshapes turbulence

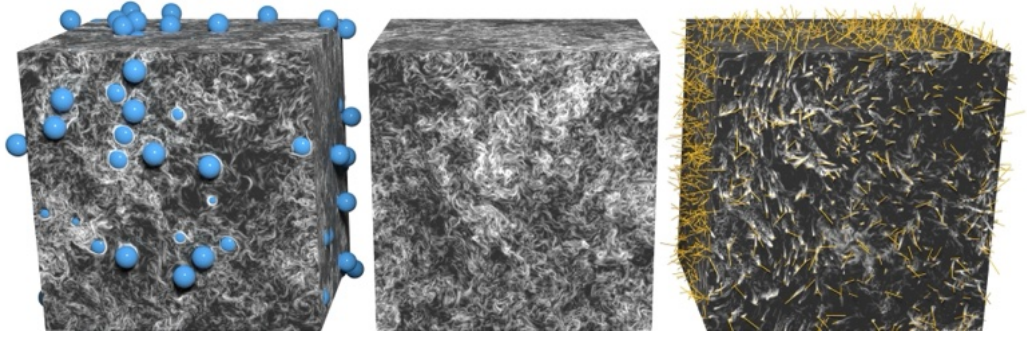
APS DFD 2025, November 25, Houston, Texas

Ianto Cannon[1,2], Stefano Olivieri[1,3], Marco E. Rosti[1]

[1]Okinawa Institute of Science and Technology, Japan

[2]University of Bremen, Germany

[3]University of Genoa, Italy



How particle shape reshapes turbulence

APS DFD 2025, November 25, Houston, Texas

Ianto Cannon[1,2], Stefano Olivieri[1,3], Marco E. Rosti[1]

[1]Okinawa Institute of Science and Technology, Japan

[2]University of Bremen, Germany

[3]University of Genoa, Italy



Motivation: Particle shapes in turbulent flows

Sandstorms

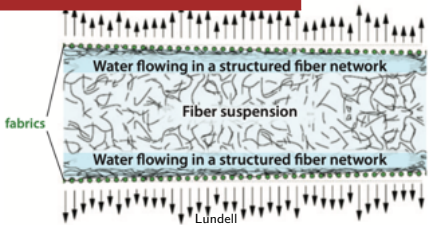


Ocean plastics



How does particle shape affect turbulent flows?





Background

Previous studies (Voth & Soldati *AFM* 2017, Brandt & Coletti *AFM* 2021, Marchioli, Rosti & Verhille *AFM* 2025):

1. Small particles ($c \ll \eta$)
2. Spherical particles
3. Limited turbulence intensity

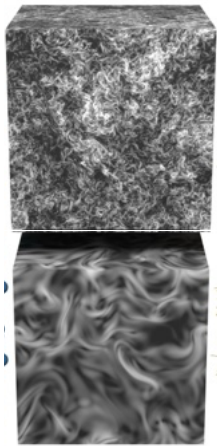
Our approach:

1. Finite size ($\eta < c < \mathcal{L}$)
2. Spheres *and* fibres
3. Wide range of turbulence intensities.

Homogeneous isotropic turbulence

- Direct numerical simulations on 1024^3 grid
- Domain size $(2\pi L)^3$
- ABC forcing (Libin & Sivashinsky *QAM* 1990)
- Five turbulence intensities

$$Re_{ABC} \equiv \frac{\sqrt{FL^3}}{\nu} = \{55.9, 112, 224, 447, 894\}.$$



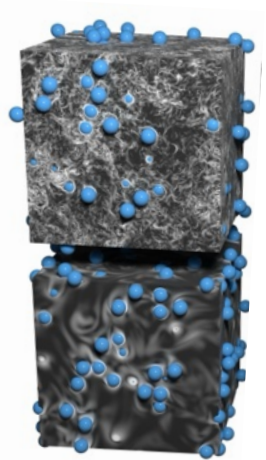
Strongly turbulent
$Re_{\mathrm{ABC}} = 894$

Weakly turbulent
$Re_{\mathrm{ABC}} = 55.9$

Spheres

- Coupled to the flow using an immersed boundary method (Hori Rosti & Takagi *CAF* 2022)
- Number: 300
- Diameter: $c = 0.5L$
- Volume fraction: 8%
- Total surface area: $230L^2$
- Six mass fractions

$$M = \frac{M_{\text{particles}}}{M_{\text{particles}} + M_{\text{fluid}}} = \{0, 0.1, 0.3, 0.6, 0.9, 1\}.$$



$Re_{\text{ABC}} = 894$

$Re_{\text{ABC}} = 55.9$

Fibres

- Coupled to the flow using an immersed boundary method (Huang, Shin & Sung *JCP* 2007)
- Number: 10^4
- Length $c = 0.5L$
- Thickness $d = 0.018L$
- Volume fraction: 0.05%
- Total surface area $230L^2$
- Six mass fractions

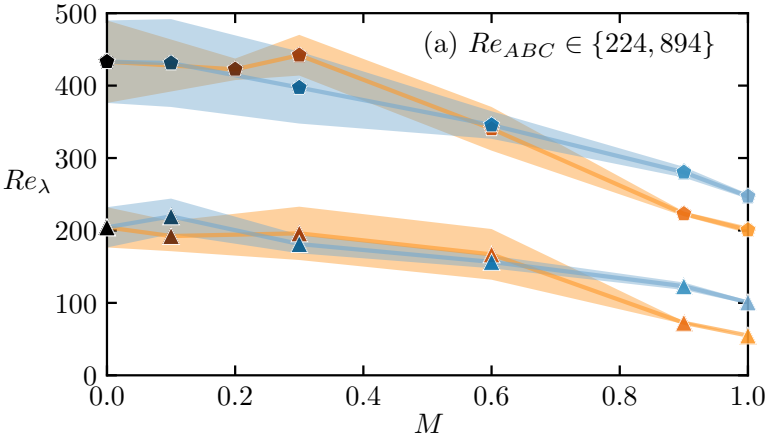
$$M = \frac{M_{\text{particles}}}{M_{\text{particles}} + M_{\text{fluid}}} = \{0, 0.1, 0.3, 0.6, 0.9, 1\}.$$



$Re_{\text{ABC}} = 894$

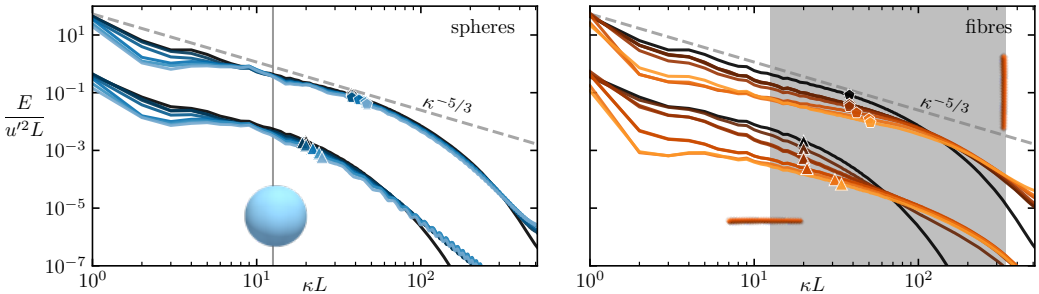
$Re_{\text{ABC}} = 55.9$

Effect on turbulence intensity



- Spheres and fibres both attenuate turbulence
- Denser particles have more effect
- Occurs at all Reynolds numbers Re_{ABC}.

Energy spectra



- The inertial range increases with the Reynolds number
- Spheres: energy reduction at $\kappa < \kappa_c$ (large scales) across all scales
- Fibres: broadband energy modulation
- Both: enhanced dissipation in viscous range.

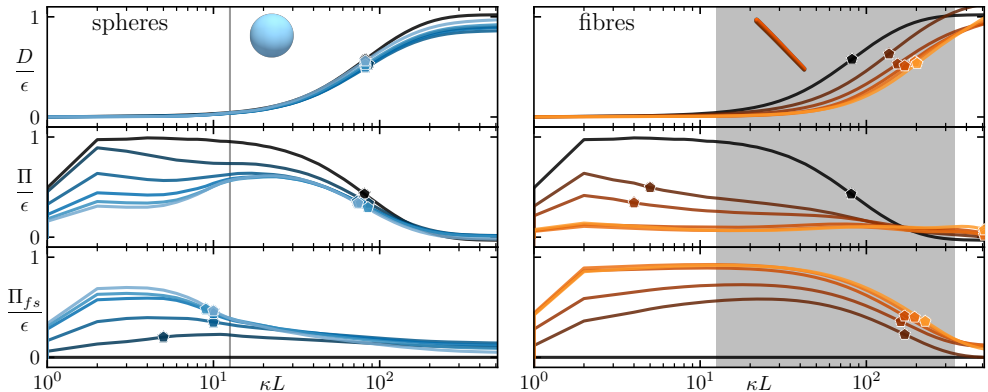
Spectral energy transfer

Energy balance: $\mathcal{F}_{inj}(\kappa) + \Pi(\kappa) + \Pi_{fs}(\kappa) + D(\kappa) = \epsilon$

dissipation

advection

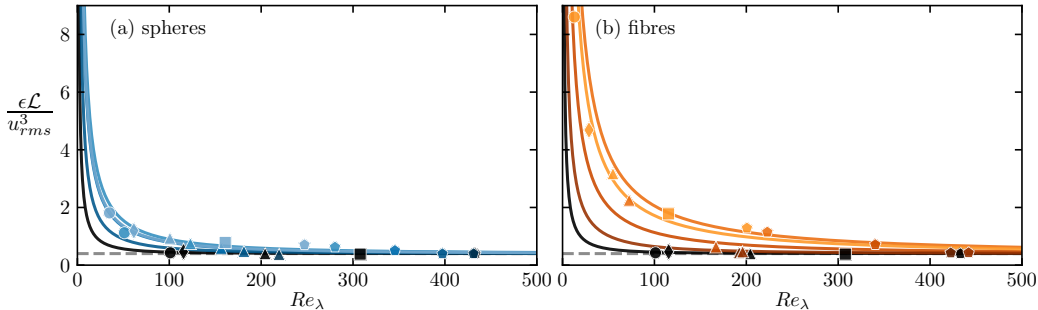
fluid-solid coupling



- Fluid-solid coupling Π_{fs} bypasses the classical cascade Π (Lucci, Ferrante & Elghobashi *JFM* 2010).
- Spheres inject at κ_c (diameter)
- Fibres inject at $\kappa \sim \kappa_d$ (thickness).

Anomalous dissipation

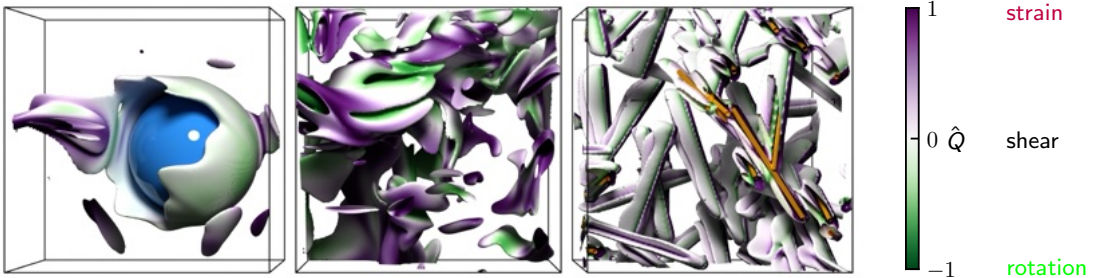
Dissipative anomaly: $\epsilon\mathcal{L}/u_{rms}^3 \to$ constant as $Re_\lambda \to \infty$ (Taylor *RSL* 1935)



- All flows converge to ≈ 0.4 (Donzis, Sreenivasan & Yeung *JFM* 2005)
- Higher mass of particles gives a slower rate of convergence
- Fibres halve the rate of convergence compared to spheres.

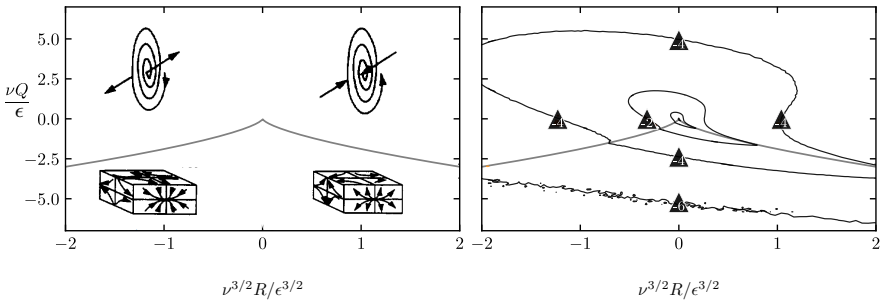
Particle wakes

Surfaces where dissipation is 6ϵ, coloured by flow topology parameter $\hat{Q} = \frac{2s_{ij}s_{ij} - \omega_i\omega_i}{2s_{ij}s_{ij} + \omega_i\omega_i}$



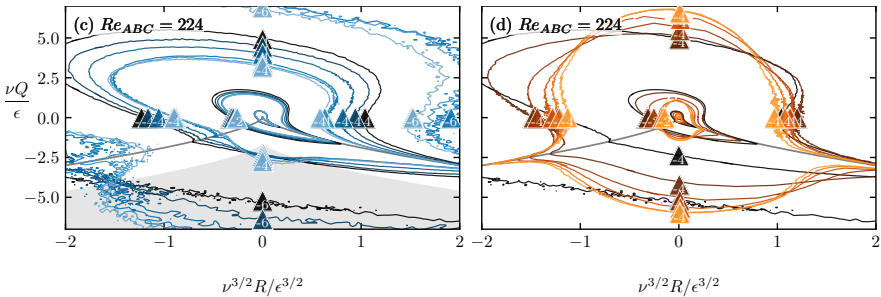
- Single phase: complex structures (Meneveau & Sreenivasan *JFM* 1991)
- Spheres: sheet-like (2D) with shear
- Fibres: tube-like (1D) structures with strain upstream, shear in wakes.
- The spatial intermittency of the dissipation is measured in Cannon, Olivieri & Rosti *PRF* 2024.

Flow topology: Q–R joint probability distribution



- Invariants of the velocity gradient tensor:
  - $Q = \frac{1}{4}\omega_i\omega_i - \frac{1}{2}s_{ij}s_{ij}$
  - $R = \frac{1}{4}\omega_i s_{ij}\omega_j - \frac{1}{3}s_{ij}s_{jk}s_{ki}$
- Single phase flow is dominated by **vortex streching** and **uniaxial strain** (Cheng & Cantwell *IAA* 1996).

Flow topology with particles



- Spheres dampen vortices
- Spheres enhance strain

- Fibres decorrelate Q and R
- Fibres greatly enhance strain.

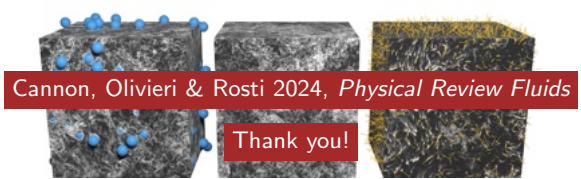
Conclusion

Findings:

1. Particles reduce turbulence intensity at all Re
2. Anomalous dissipation: $\epsilon \mathcal{L}/u_{rms}^3 \to 0.4$ as $Re_\lambda \to \infty$
3. Particles bypass the classical turbulent energy cascade
   - Spheres inject at diameter scale
   - Fibres inject at thickness scale
4. Different dissipation structures:
   - Spheres: 2D sheets
   - Fibres: 1D tubes

Implication:

- For pollen dispersal, ocean plastics, sandstorms and papermaking, **particle shape matters, not just mass**



Cannon, Olivieri & Rosti 2024, *Physical Review Fluids*

Thank you!

Bibliography I

Ashurst, Wm. T. et al. (Aug. 1987). "Alignment of Vorticity and Scalar Gradient with Strain Rate in Simulated Navier–Stokes Turbulence". In: *The Physics of Fluids* 30.8, pp. 2343–2353. ISSN: 0031-9171. DOI: 10.1063/1.866513. (Visited on 08/05/2023).

Donzis, D. A., K. R. Sreenivasan, and P. K. Yeung (June 2005). "Scalar Dissipation Rate and Dissipative Anomaly in Isotropic Turbulence". In: *Journal of Fluid Mechanics* 532, pp. 199–216. ISSN: 0022-1120, 1469-7645. DOI: 10.1017/S0022112005004039. (Visited on 08/12/2022).

Eswaran, V. and S. B. Pope (Jan. 1988). "An Examination of Forcing in Direct Numerical Simulations of Turbulence". In: *Computers & Fluids* 16.3, pp. 257–278. ISSN: 0045-7930. DOI: 10.1016/0045-7930(88)90013-8. (Visited on 04/22/2020).

Peng, Cheng, Qichao Sun, and Lian-Ping Wang (May 2023). "Parameterization of Turbulence Modulation by Finite-Size Solid Particles in Forced Homogeneous Isotropic Turbulence". In: *Journal of Fluid Mechanics* 963, A6. ISSN: 0022-1120, 1469-7645. DOI: 10.1017/jfm.2023.317. (Visited on 05/13/2023).

Shaik, Subhani and René Van Hout (Jan. 2023). "Kinematics of Rigid Fibers in a Turbulent Channel Flow". In: *International Journal of Multiphase Flow* 158, p. 104262. ISSN: 03019322. DOI: 10.1016/j.ijmultiphaseflow.2022.104262. (Visited on 08/06/2023).

Snook, Braden, Elisabeth Guazzelli, and Jason E. Butler (Dec. 2012). "Vorticity Alignment of Rigid Fibers in an Oscillatory Shear Flow: Role of Confinement". In: *Physics of Fluids* 24.12, p. 121702. ISSN: 1070-6631. DOI: 10.1063/1.4770141. (Visited on 05/17/2023).